

Composition Pair B1.06 + B1.14 — The Enabling Pair Where the DNA Layer Provides the Governed Substrate That Directed Selection Modifies, and Directed Selection Provides the Governance Mechanism That Gives the DNA Layer Its Evolution Significance as a Continuously Improving Governed Specification

A derivation note from the Coordination Knowledge Substrate (CKS) theory series — Series B, Phase B4, Note B4.19.

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in The Instinct/Reasoning Separation Outside the Model (Li, April 2026), the second paper in the CKS theory series following Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026).

Abstract

Two architectural commitments in Paper 2 of the Coordination Knowledge Substrate (CKS) theory series stand in an enabling relationship that is architecturally necessary rather than incidental. B1.06 — the two-layer structure distinguishing the DNA layer (stabilized orchestration substrates) from the action layer (operational records) within every cell — provides the architecturally distinct, separately maintained substrate that directed selection per B1.14 requires as its governance target. B1.14 — directed selection, the human-governed mechanism by which DNA layer content is modified through Paper 1's standard authority architecture — provides the governance process that gives the DNA layer its significance as an evolving specification rather than a frozen deployment artifact. Neither commitment is architecturally complete without the other. This note formalizes the enabling pair: identifies the two enabling directions, articulates the mutual dependency, names the DNA version chain per B2.69 as the operational embodiment of this pair's composition, distinguishes the DNA layer target from the action layer in order to bound the Layer Conflation anti-pattern per B3.07, and provides co-presence requirements, detection criteria, and failure modes for deployments where one commitment is present without the other.

1. Why this pair requires joint formalization

Paper 2 introduces both B1.06 and B1.14 as individually defensible architectural commitments. B1.06 (Paper 2 §5.3) establishes that every cell contains two architecturally distinct layers: a DNA layer holding stabilized orchestration substrates — behavior substrates, harness substrates, schemas, lifecycle policies — that constitute the authoritative governing content for entity behavior; and an action layer holding operational records, task instances, and outputs that constitute the cell's lived history. B1.14 (Paper 2 §7.2) establishes that DNA evolution operates through directed selection: human-governed modification of DNA layer content under explicit goals, authorized through Paper 1's standard authority architecture, with selection criteria that are themselves substrate content governed by the same authority.

The purpose of a composition pair formalization — following the structure established in B4.01 — is to make explicit an enabling relationship that is not derivable from reading either commitment in isolation. In this case, the relationship is constitutive: B1.06 provides directed selection with the architectural substrate it requires to be operationally defined, and B1.14 provides the DNA layer with the governance dynamics that make it architecturally significant as more than a static specification. These are not incidental connections. They are the reason each commitment is in the architecture at all.

2. The enabling direction: B1.06 provides B1.14 its architectural substrate

Directed selection per B1.14 is defined as human-governed modification of DNA content. That definition has an implicit referent: "DNA content" must denote a specific, separately maintained architectural layer — not the LLM's internal state, not the action layer's operational records, not an informal reference to "the rules governing this entity." Without B1.06, the phrase "modify the DNA" lacks an architectural substrate referent.

B1.06 provides that referent. The DNA layer per B1.06 is architecturally distinct from the action layer: it is separately maintained, it is the authoritative source of governing specifications per A1.08 substrate-as-source-of-truth, and it is what governs entity behavior at every point in the cell's lifecycle. The DNA layer's contents are specifically orchestration substrates — behavior substrates specifying how the cell acts, harness substrates specifying how it interacts, schemas specifying what it records, lifecycle policies specifying how it changes. These are the objects that directed selection modifies.

Without B1.06, directed selection would be operating on "the specification" in the abstract — a conceptual target without an architectural implementation. The governance acts that Paper 1's standard authority architecture enables — the inspect right (A2.47), the modify right (A2.04), the override right (A2.40) — require a definite substrate to operate on. B1.06 supplies the definite substrate. It gives directed selection its target, its boundary, and its authority claim: the DNA layer is the authoritative governing specification, so modifying it through governed process is what produces authoritative behavioral change in the entity.

The enabling direction is therefore precise: **B1.06 enables B1.14 by providing the architecturally distinct, separately maintained, authoritative substrate layer that directed selection requires as its governance target.** Without the DNA layer as an architectural reality, directed selection has no operationally defined object.

3. The significance direction: B1.14 gives B1.06 its evolution significance

The DNA layer per B1.06 could, in principle, exist as a static specification. Set at deployment, encoding the governing substrates that authorize cell behavior, satisfying A1.08's substrate-as-source-of-truth commitment for the deployment lifetime without modification. A static DNA layer would be architecturally coherent. It would govern entity behavior. It would support path retraceability per A1.07. It would fulfill the commitments B1.06 makes.

But a static DNA layer cannot improve through governance. It cannot be refined in response to operational experience. It cannot incorporate lessons from the action layer's accumulated evidence about what works and what does not. It cannot respond to changes in the environment that the governing humans can observe but that were not anticipated at deployment time. A static DNA layer is a deployment artifact, not an evolving governed specification.

B1.14 directed selection is what converts the DNA layer from deployment artifact to governed evolving specification. Each directed selection event — authorized by the human holders of the modify right, executed as a DNA layer modification, recorded with provenance — produces a new version of the DNA layer. The new version governs forward from the point of modification. The version history is the record of every governance decision that has shaped entity behavior over time.

This is the significance direction: **B1.14 gives B1.06 evolution significance by providing the governance process through which the DNA layer improves over the cell's operational lifetime.** Without directed selection, the DNA layer is frozen at its deployment version. With directed selection, it is a continuously improving governed specification — the architectural expression of governance's ability to learn.

The asymmetry between the two directions is worth naming. The enabling direction (B1.06 → B1.14) is primarily architectural: B1.06 is required before B1.14 can be operationally defined. The significance direction (B1.14 → B1.06) is primarily dynamic: B1.14 is what gives B1.06 a role in the cell's ongoing evolution rather than only in its initial configuration. Both directions are real, and both are required for a complete account of why this pair is in the architecture together.

4. Mutual dependency: each commitment requires the other to be architecturally complete

The two enabling directions jointly produce a mutual dependency. Neither B1.06 nor B1.14 is architecturally complete without the other's presence — not in the sense that either is logically incoherent in isolation, but in the sense that each achieves its full architectural purpose only in

composition.

B1.06 requires B1.14 to achieve its purpose as a governed substrate. The DNA layer is architecturally distinct and separately maintained so that it can be the authoritative target of governance acts. If no governance acts ever modify it — if directed selection per B1.14 is absent — the DNA layer serves as a governance target in name without being one in practice. Its design as a separately maintained authoritative substrate implies a modification process; B1.14 is that process. Without it, the DNA layer's architectural separation achieves nothing that a simpler static specification would not also achieve.

B1.14 requires B1.06 to achieve its purpose as a governed evolution mechanism. Directed selection is defined as modification of DNA content. If the DNA layer per B1.06 is absent — if there is no architecturally distinct substrate whose contents are the authoritative governing specifications — then directed selection is operating on an undefined target. The governance acts authorized by Paper 1's standard authority architecture (inspect per A2.47, modify per A2.04, record per A2.40) require a substrate to operate on; without B1.06, these acts have no architectural ground.

The mutual dependency produces the composition relationship this note formalizes: the DNA layer per B1.06 is the governed substrate, and directed selection per B1.14 is the governance process that evolves it. Together they produce what neither produces alone: a governed substrate that improves through human governance decisions over a cell's operational lifetime.

5. The DNA version chain as operational embodiment

The DNA version chain per B2.69 is the operational product of this pair's composition. It is not merely a useful feature of deployments that instantiate both B1.06 and B1.14; it is the direct consequence of their interaction over time, and its structure reflects both commitments fully.

Each element of the DNA version chain is a complete DNA layer state per B1.06: a snapshot of the orchestration substrates at a point in time, sufficient to govern entity behavior for the period it was in force. Each transition between consecutive elements is a directed selection event per B1.14: a human-governed modification authorized through Paper 1's standard authority architecture, recorded with provenance per A2.40. The full chain is the DNA layer's governed evolution history — every behavioral specification the entity has ever operated under, every governance decision that modified it, every moment of transition from one governing state to the next.

The DNA version chain has three architectural properties that make it the embodiment of this pair. *First*, it makes the enabling direction visible. The chain consists of DNA layer states, confirming that B1.06's architectural substrate is what directed selection has been modifying. If the transitions in the chain were found to modify action layer content rather than DNA layer content, that would be evidence of Layer Conflation per B3.07 — but the chain's structure, by design, records DNA layer states only.

Second, it makes the significance direction visible. The existence of a version chain with more than one element is direct evidence that B1.14 has given the DNA layer evolution significance: the layer has changed from its deployment version, and it has changed through governance. A DNA layer that has never been modified produces a version chain of length one — the deployment version — which is the signature of Evolution Stasis per B3.24.

Third, it provides the provenance path that makes directed selection auditable. Per A2.40, each transition is recorded. The record includes what changed, who authorized the change, under what authority, and at what time. The DNA version chain is therefore not merely a sequence of states; it is a governed history of how human authority has shaped entity behavior over time. This is the operational expression of the composition pair at work.

6. The action layer contrast and the Layer Conflation anti-pattern

B1.06 introduces two layers, and the pair B1.06 + B1.14 operates on one of them. Precision about which layer directed selection targets is essential, because confusing the two is the Layer Conflation anti-pattern per B3.07.

The DNA layer contains orchestration substrates — the governing specifications that authorize and direct entity behavior. It is what directed selection modifies. The action layer contains operational records — task instances, outputs, the lived history of what the cell has done. It is what action-feedback evolution operates on, but it is not what directed selection modifies. The action layer's content is produced by the cell's operation; the DNA layer's content governs that operation.

The contrast matters architecturally because the two layers carry different authority claims. DNA layer content is authoritative governing specification per A1.08: it is the source of truth for how the entity is supposed to behave. Action layer content is authoritative operational record: it is the source of truth for what the entity has done. Modifying action layer content through a directed selection event would be modifying the historical record, not the governing specification. That is not what directed selection does, and an implementation that conflates the two violates B1.06's architectural separation.

Layer Conflation per B3.07 arises when a deployment treats action layer records as the target of directed selection, or equivalently, when DNA layer specifications are stored in the action layer's position in the architecture. Both variants produce the same failure: directed selection's governance acts are applied to the wrong layer. The governance signal modifies the historical record while leaving the governing specification unchanged, or modifies the governing specification without maintaining it as a separately authoritative layer. Either way, the A1.08 authority claim the DNA layer carries is undermined.

B1.06's two-layer clarity is the architectural protection against this failure. By maintaining the DNA layer and action layer as architecturally distinct — separately maintained, separately addressed, carrying distinct authority claims — B1.06 provides the structural condition under which directed

selection can operate on the correct target. The pair's composition requires this clarity: B1.14's directed selection presupposes the architectural separation B1.06 establishes.

7. Co-presence requirements, operational integration, detection, and failure modes

Co-presence requirements. The full composition pair requires both commitments to be present. Two degraded configurations are possible.

B1.06 without B1.14 — two-layer structure without directed selection. The DNA layer exists and governs entity behavior. The cell can operate. The A1.08 authority claim is fulfilled at deployment scope. But the DNA layer cannot improve through governance: no directed selection events are authorized, no new versions are created, and the entity's governing specification is frozen at its deployment state. This is the substrate for Evolution Stasis per B3.24. The cell's behavior can change only through instinct evolution (LLM and infrastructure upgrades) or action-feedback evolution (scope reduction and DNA refactoring from usage patterns), but not through the intentional, goal-directed modification that B1.14 enables. The two-layer structure is present; its evolution capacity is not.

B1.14 without B1.06 — directed selection without a two-layer substrate. Governance acts are authorized and executed, and modifications to "the rules" are recorded. But the modifications do not land on an architecturally distinct DNA layer that holds its authority claim per A1.08. The governed modification process is operational, but it is operating on a substrate that lacks the architectural separation that makes DNA layer modifications authoritative. The result is governance activity without the substrate structure that gives governance activity its architectural effect.

Full co-presence — both commitments present. The DNA layer per B1.06 is the governed substrate. Directed selection per B1.14 is the governance process that evolves it. Together they produce continuously improving governed entity behavior specifications, with the DNA version chain per B2.69 as the operational record of that improvement.

Operational integration. The composition pair operates through the following cycle during a cell's deployment lifetime: (1) the DNA layer per B1.06, containing orchestration substrates established at deployment, governs entity behavior; (2) operational experience accumulates in the action layer; (3) a directed selection event per B1.14 is authorized by the human holders of the modify right per A2.04, executed as a modification to specific DNA layer content, and recorded with provenance per A2.40; (4) a new DNA version is created per B2.69, timestamped and linked to its predecessor; (5) the new DNA version governs entity behavior forward per the updated substrates, which remain authoritative per A1.08; (6) the cycle continues through the cell's operational lifetime, producing the DNA version chain.

Detection. Three detection criteria confirm this pair is properly instantiated. *B2.72 directed selection verification*: are all DNA modifications authorized through Paper 1's standard authority architecture and recorded with provenance? *B2.69 DNA version chain*: does a version chain exist

documenting the DNA layer's governed evolution from its deployment version? *Pair completeness*: does directed selection specifically target DNA layer content, as confirmed by the version chain, and not action layer content per B3.07?

Failure modes. Three failure modes are associated with this pair. *Evolution Stasis per B3.24*: B1.06 is present but B1.14 is absent; the DNA layer exists but is never modified through directed selection, freezing the entity's governing specification at deployment. *Layer Conflation per B3.07*: B1.06's layer distinction is violated; directed selection targets action layer content rather than DNA layer content, or DNA layer content is stored in the action layer's architectural position, undermining the A1.08 authority claim the DNA layer carries. *Ungoverned DNA Modification per B3.15*: B1.14's governance process is bypassed; DNA layer content is modified without authorization through Paper 1's standard authority architecture, without provenance recording, or by actors without the modify right — producing changes to the governing specification that carry no authority claim and cannot be traced.

8. Operational test

A deployment instantiates the enabling pair B1.06 + B1.14 fully if and only if all of the following are true at any moment during the cell's operational lifetime:

1. There exists an architecturally distinct DNA layer containing orchestration substrates that are maintained as the authoritative governing specification for the cell per A1.08 — separate from the action layer, separately addressed, separately maintained.
2. There exists a directed selection process by which DNA layer content can be modified, where each modification is authorized through Paper 1's standard authority architecture (inspect right per A2.47, modify right per A2.04, record per A2.40), executed by the human holders of the modify right, and recorded with provenance linking the new DNA version to its predecessor.
3. The DNA version chain per B2.69 documents the DNA layer's governed evolution from its deployment version, with each transition corresponding to a traceable directed selection event — not to an undirected modification, not to action layer content, and not to ungoverned change.
4. Directed selection targets DNA layer content specifically — not action layer operational records — as confirmed by the version chain's structure and the distinct authority claims the two layers carry per B1.06.

A deployment that satisfies all four criteria instantiates both the enabling direction (B1.06 provides B1.14 its architectural substrate) and the significance direction (B1.14 gives B1.06 its evolution significance). A deployment that satisfies criterion 1 but not 2 or 3 is in the Evolution Stasis configuration. A deployment that satisfies criteria 2 and 3 but not 4 has a Layer Conflation violation. A deployment that satisfies criteria 1 and 4 but where criterion 2's governance process is bypassed has an Ungoverned DNA Modification violation.

Source

Li, Wenxin. *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. See especially §5.3 (DNA/action layer distinction), §7.2 (directed selection and DNA evolution), and §8 (multi-shaped governance and directed selection authority architecture).

Li, Wenxin. *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. See §2.3 (authority architecture, inspect/modify/override rights), §3.1 (path retraceability and provenance), §11.3 (substrate as source of truth).

Self-citation context

This note is the nineteenth composition pair formalization in Phase B4 of the CKS derivation note series (Series B, Paper 2 derivations). It formalizes the enabling relationship between B1.06 (two-layer structure: DNA layer and action layer) and B1.14 (directed selection: human-governed DNA modification). Related notes in this series include the B1.06 foundational note (two-layer structure), the B1.14 foundational note (directed selection), B2.25 (DNA layer specification), B2.69 (DNA version management), B2.72 (directed selection verification), B3.07 (Layer Conflation anti-pattern), B3.15 (Ungoverned DNA Modification anti-pattern), and B3.24 (Evolution Stasis anti-pattern).